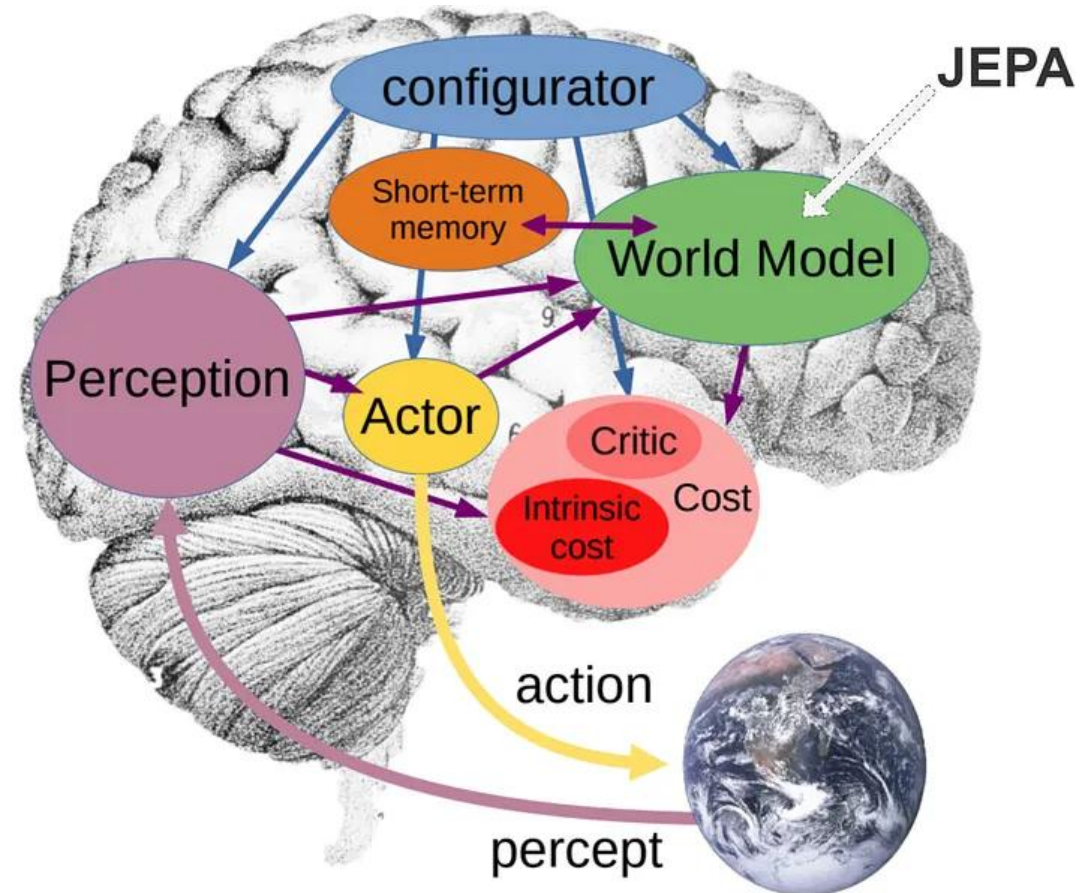


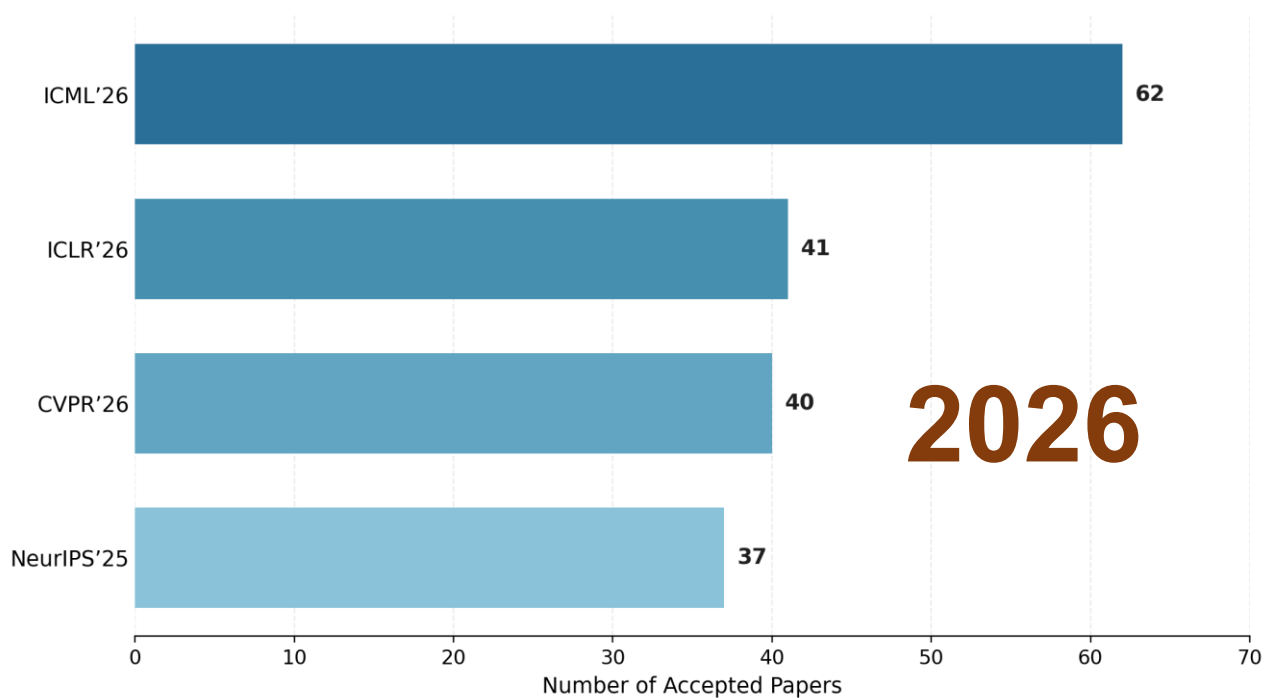
World Model

Revolution or evolution ?

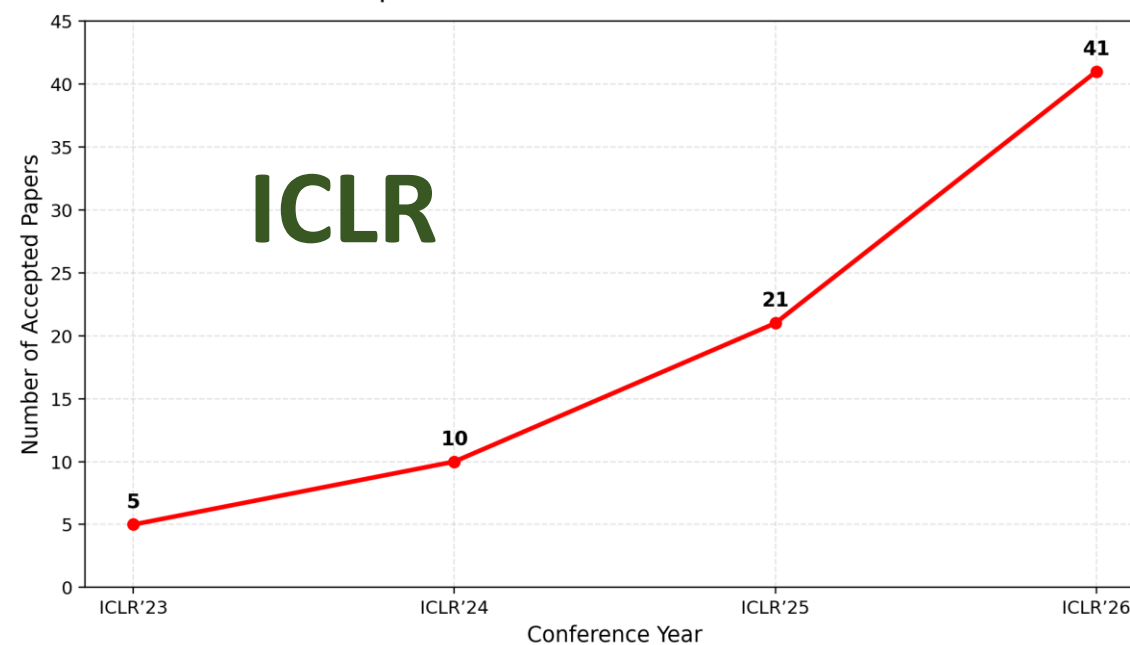


Everyone Is Building a World Model Now (:

Accepted Papers Related to World Models



ICLR Papers Related to 'World Model' Over Time



Drawbacks of LLMs



Hallucination



Weak world
understanding



Limited reasoning



Poor planning



Data hungry

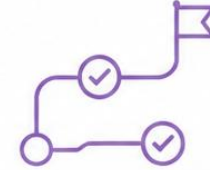
World Model Benefits



**Grounded
understanding**



**Better
reasoning**



**Better
planning**

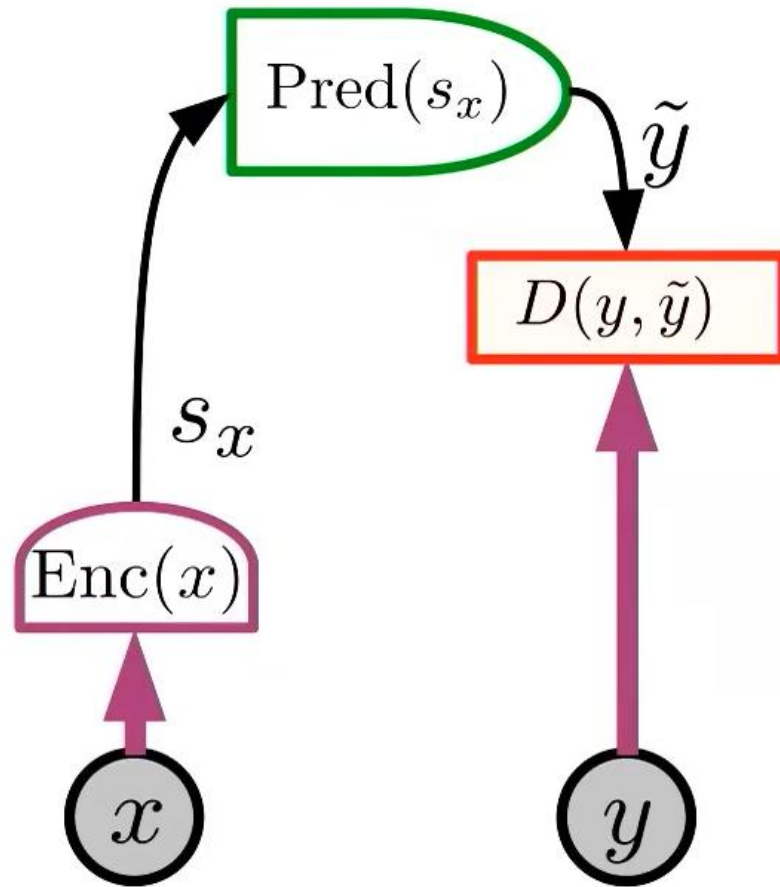


**Fewer
hallucinations**

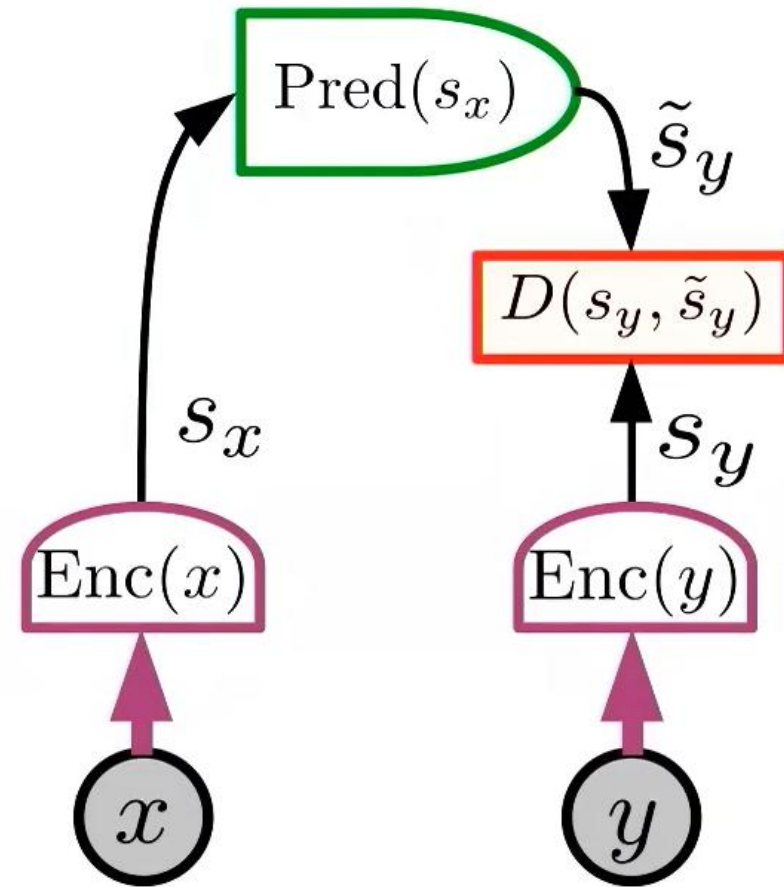


**More data
efficient**

Architectures: Generative vs Joint Embedding

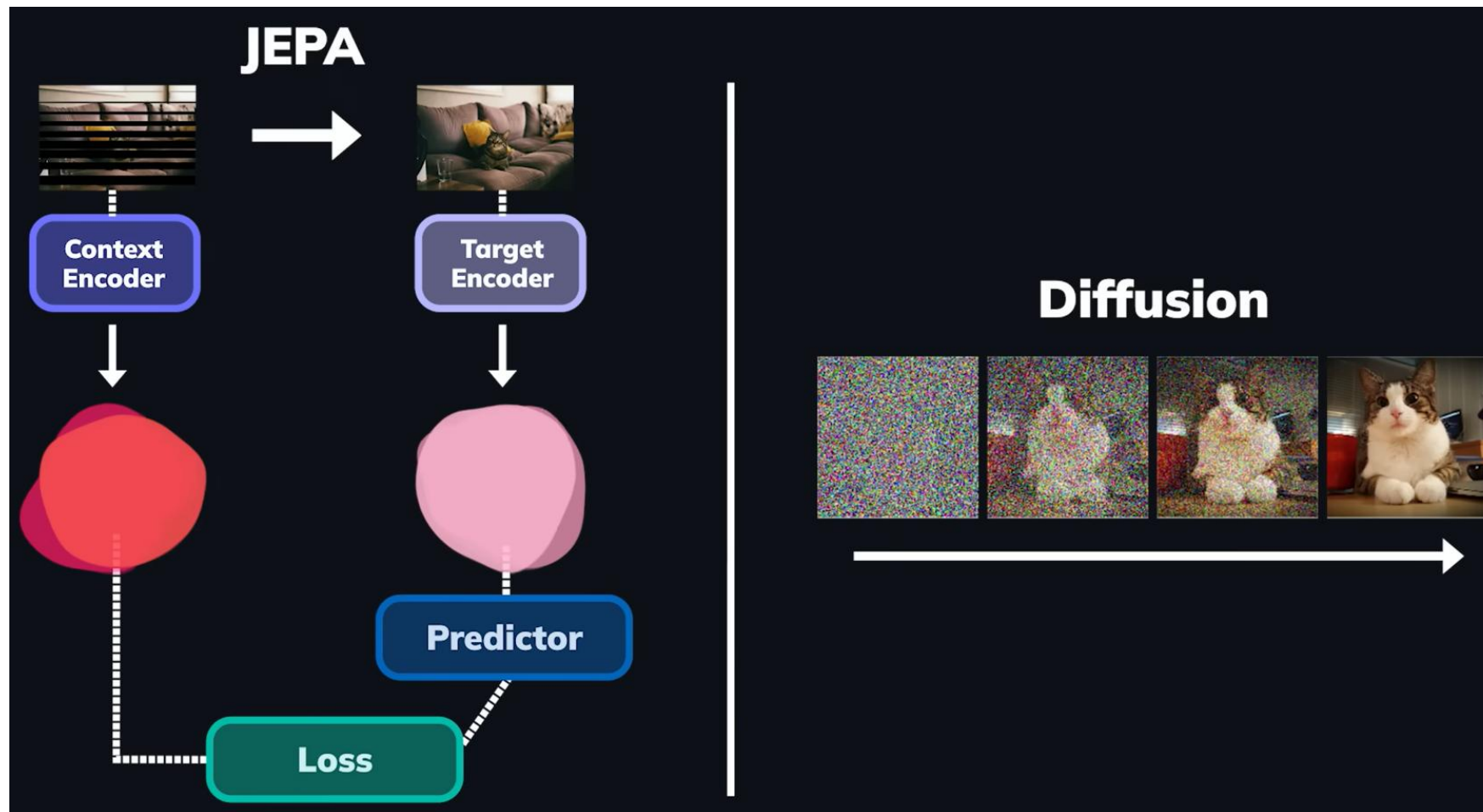


a) Generative Architecture
Examples: VAE, MAE...

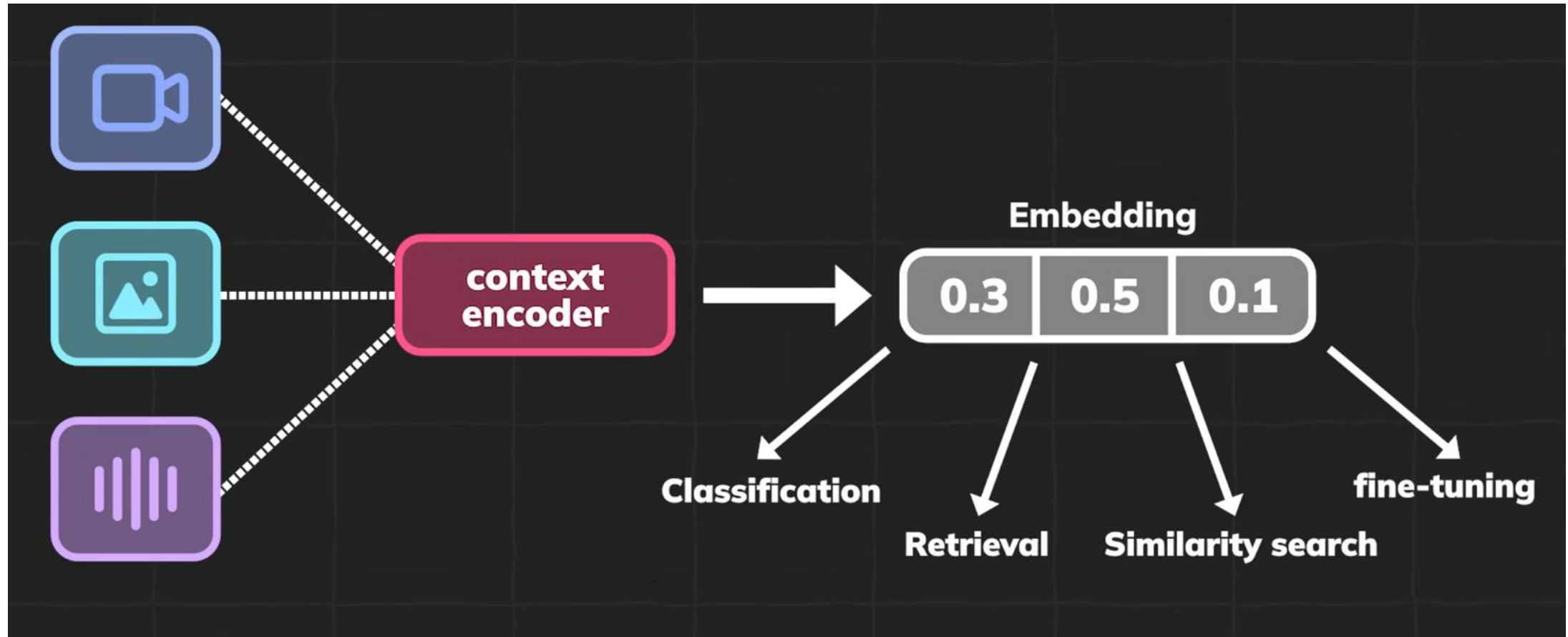


b) Joint Embedding Architecture

Example

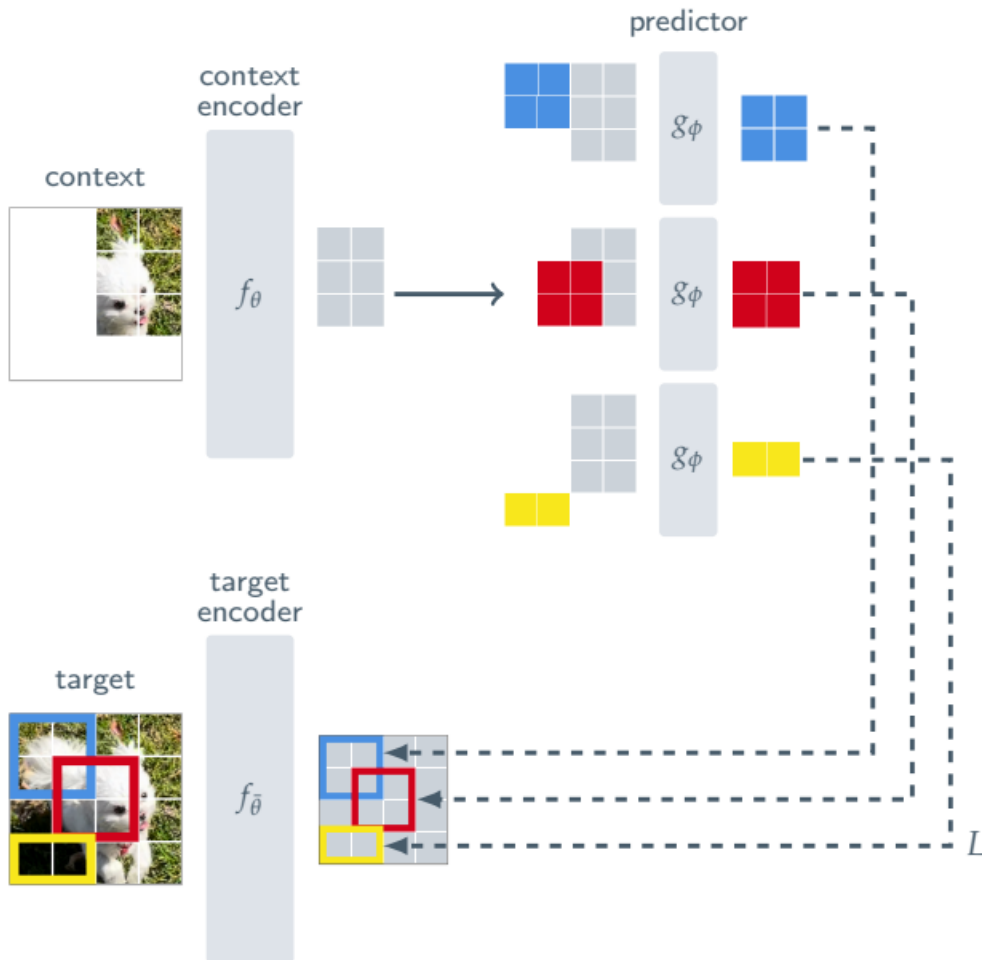


Representation Extraction

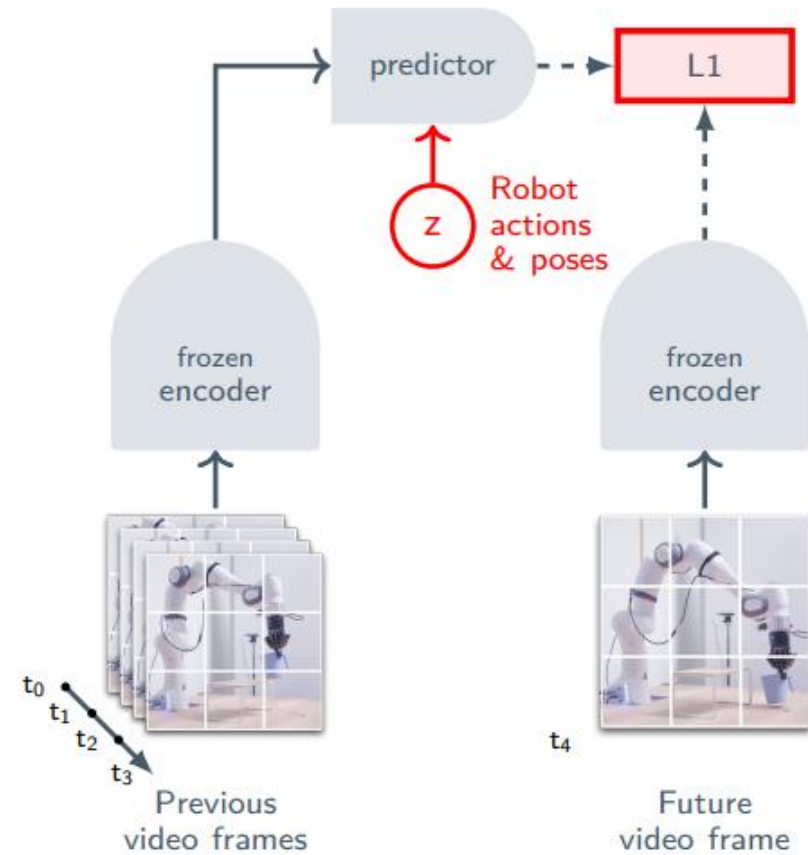


Variants of JEPA

I-JEPA



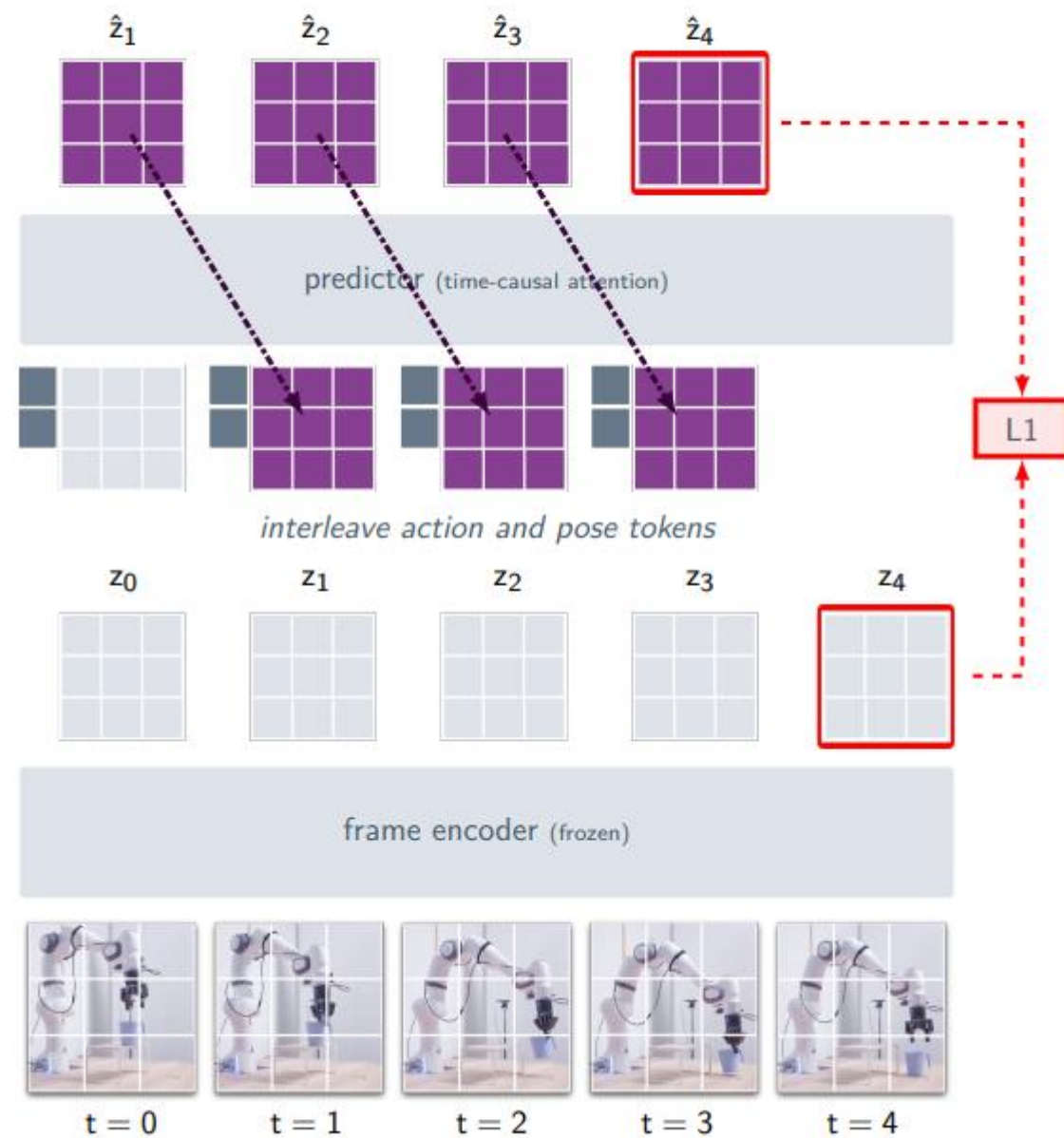
V-JEPA 2-AC



VJEPA 2-AC Teacher Forcing Loss



VJEPA 2-AC Rollout Loss



Representation Collapse

